Algebraic and geometric multiplicity of eigenvalues

by Marco Taboga, PhD

The algebraic multiplicity of an eigenvalue is the number of times it appears as a root of the characteristic polynomial (i.e., the polynomial whose roots are the eigenvalues of a matrix).

The geometric multiplicity of an eigenvalue is the dimension of the linear space of its associated eigenvectors (i.e., its eigenspace).

In this lecture we provide rigorous definitions of the two concepts of algebraic and geometric multiplicity and we prove some useful facts about them.

Looking for a **geometric multiplicity calculator** or a **step-by-step tutorial** on how to calculate the geometric multiplicity? Follow this link.

Algebraic multiplicity

Let us start with a definition.

Definition Let be a matrix. Denote by the possibly repeated eigenvalues of , which solve the characteristic equation

We say that an eigenvalue has algebraic multiplicity if and only if there are no more and no less than solutions of the characteristic equation equal to .

Let us see some examples.

Example Consider the matrix

The characteristic polynomial is

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The roots of the polynomial, that is, the solutions of are

Thus, has two distinct eigenvalues. Their algebraic multiplicities are

because they are not repeated.

Example Define the matrix

Its characteristic polynomial is

The roots of the polynomial, that is, the solutions of are

Thus, has one repeated eigenvalue whose algebraic multiplicity is

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Geometric multiplicity

Recall that each eigenvalue is associated to a linear space of eigenvectors, called eigenspace.

Definition Let be a matrix. Let be one of the eigenvalues of and denote its associated eigenspace by . The dimension of is called the geometric multiplicity of the eigenvalue .

Let's now make some examples.

Definition Consider the matrix

The characteristic polynomial is

The roots of the polynomial are

The eigenvectors associated to are the vectors

that solve the equation

or

The last equation implies that

Therefore, the eigenspace of is the linear space that contains all vectors of the form

where can be any scalar. Thus, the eigenspace of is generated by a single vector

Therefore, it has dimension . As a consequence, the geometric multiplicity of is .

Example Consider the matrix

The characteristic polynomial is

$$
\begin{aligned}
f(\lambda) &= \det\left(\lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}\right) \\
&= \det\left(\begin{bmatrix} \lambda - 1 & 0 \\ 1 & \lambda - 1 \end{bmatrix}\right) \\
&= (\lambda - 1) \cdot (\lambda - 1) - 0 \cdot 1 \\
&= (\lambda - 1) \cdot (\lambda - 1)
\end{aligned}
$$

and its roots are

Thus, there is a repeated eigenvalue ( ) with algebraic multiplicity equal to 2. Its associated eigenvectors

solve the equation

$$\begin{bmatrix} \lambda_1 - 1 & 0 \\ 1 & \lambda_1 - 1 \end{bmatrix} \begin{bmatrix} x_{11} \\ x_{21} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

or

The equation is satisfied for and any value of . As a consequence, the eigenspace of is the linear space that contains all vectors of the form

where can be any scalar. Since the eigenspace of is generated by a single vector

it has dimension . As a consequence, the geometric multiplicity of is 1, less than its algebraic multiplicity, which is equal to 2.

Example Define the matrix

The characteristic polynomial is

$$\begin{aligned} f(\lambda) &= \det\left(\lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}\right) \\ &= \det\left(\begin{bmatrix} \lambda - 2 & 0 \\ 0 & \lambda - 2 \end{bmatrix}\right) \\ &= (\lambda - 2) \cdot (\lambda - 2) - 0 \cdot 0 \\ &= (\lambda - 2) \cdot (\lambda - 2) \end{aligned}$$

and its roots are

Thus, there is a repeated eigenvalue ( ) with algebraic multiplicity equal to 2. Its associated eigenvectors

solve the equation

$$\begin{bmatrix} \lambda_1 - 2 & 0 \\ 0 & \lambda_1 - 2 \end{bmatrix} \begin{bmatrix} x_{11} \\ x_{21} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

or

The equation is satisfied for any value of and . As a consequence, the eigenspace of is the linear space that contains all vectors of the form

where and are scalars that can be arbitrarily chosen. Thus, the eigenspace of is generated by the two linearly independent vectors

Hence, it has dimension . As a consequence, the geometric multiplicity of is 2, equal to its algebraic multiplicity.

A takeaway message from the previous examples is that the algebraic and geometric multiplicity of an eigenvalue do not necessarily coincide.

Relationship between algebraic and geometric multiplicity

The following proposition states an important property of multiplicities.

Proposition Let be a matrix. Let be one of the eigenvalues of . Then, the geometric multiplicity of is less than or equal to its algebraic multiplicity.

Proof

Defective eigenvalues

When the geometric multiplicity of a repeated eigenvalue is strictly less than its algebraic multiplicity, then that eigenvalue is said to be **defective**.

An eigenvalue that is not repeated has an associated eigenvector which is different from zero. Therefore, the dimension of its eigenspace is equal to 1, its geometric multiplicity is equal to 1 and equals its algebraic multiplicity. Thus, an eigenvalue that is not repeated is also non-defective.

Solved exercises

Below you can find some exercises with explained solutions.

Exercise 1

Find whether the matrix

has any defective eigenvalues.

Solution

Exercise 2

Define

Determine whether possesses any defective eigenvalues.

Solution

How to cite

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